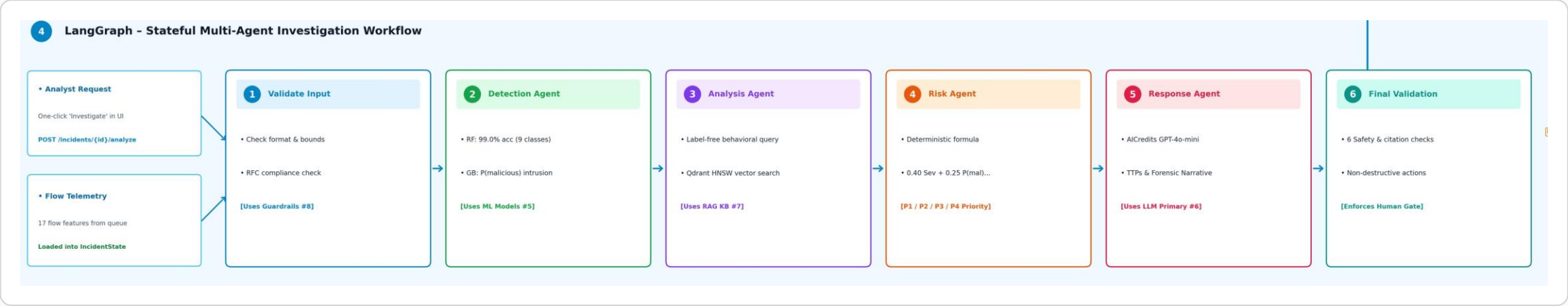
AI-Powered Cybersecurity Incident Response Assistant

Multi-agent triage, hybrid RAG and explainable risk scoring for SOC analysts.



Project review, 10 minutes

Alert volume outruns analysts

30 min

to triage one incident by hand

5 sec

target with the assistant

500 to 1,000 alerts per shift

Over 85% of triage time goes to benign noise.

No historical context

Recurring threats are investigated from scratch.

Unsafe LLM output

Hallucinated CVEs and unreviewed network changes.

Five layers, loosely coupled

Interface — React 19 single-page app and Streamlit console

API — FastAPI with JWT roles, input guardrails and trace IDs

Orchestration — LangGraph state machine with six nodes

Intelligence

ML models

Random Forest, Gradient Boosting, Isolation Forest, SHAP

Hybrid RAG

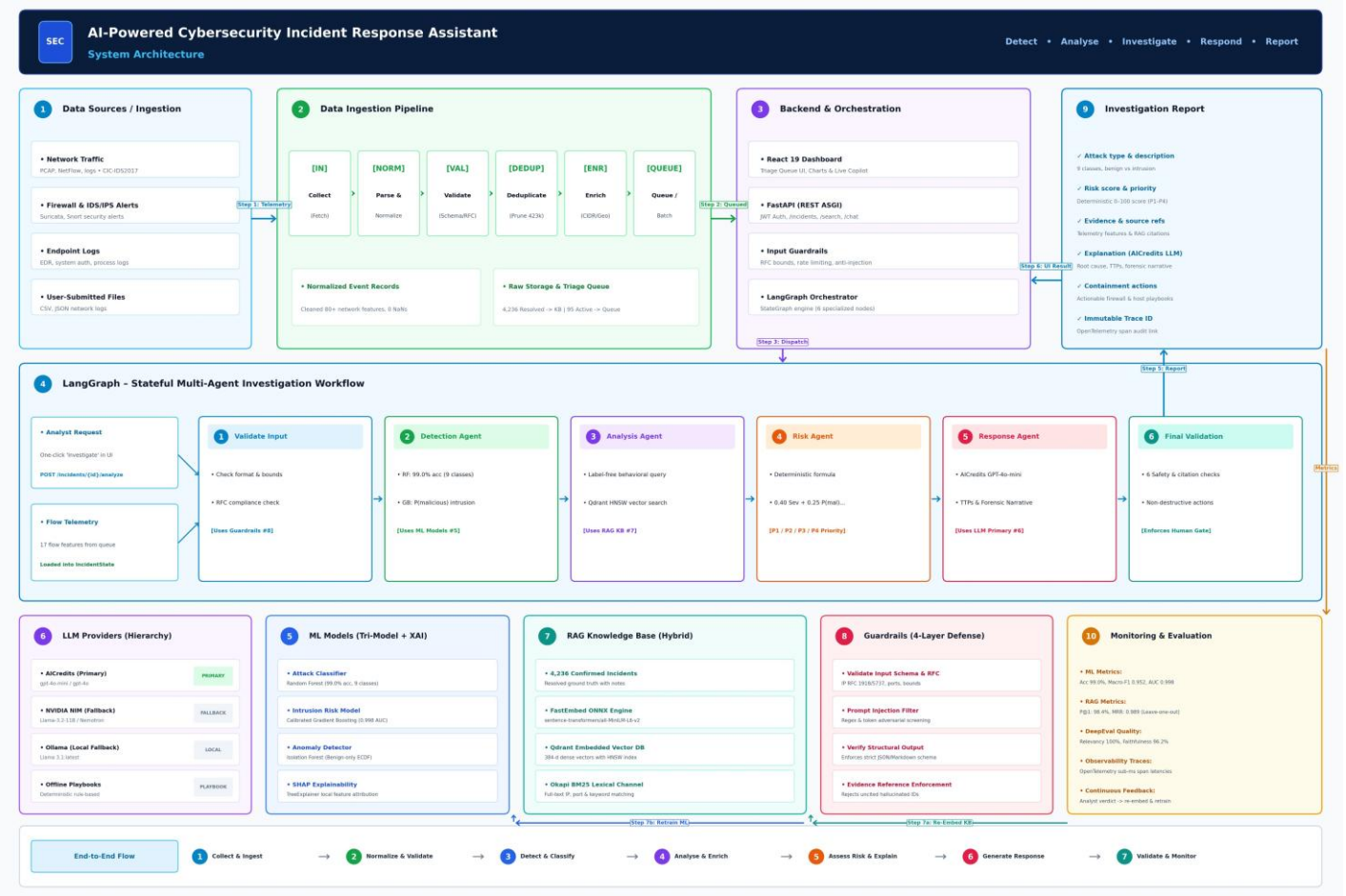
MiniLM embeddings in Qdrant, fused with BM25

LLM chain

AICredits, NVIDIA NIM, Ollama, offline playbooks

Data — SQLite or PostgreSQL system of record, embedded Qdrant index, versioned model store

The full picture



10

components in one lifecycle

Detect

Analyse

Investigate

Respond

Report

Six steps from alert to report

Validate

Schema, IP and port bounds, injection screen

Detect

Attack family, intrusion and anomaly scores, SHAP drivers

Analyse

Hybrid search across 4,236 past incidents

Score

Weighted 0 to 100 risk, priority P1 to P4

Respond

LLM drafts the report and containment playbook

Verify

Six output checks, analyst approval gate

In

Analyst clicks Investigate

Shared state

One IncidentState carries evidence between nodes

Out

Report, risk priority and a trace ID

What powers each decision

Detect

99.0%

accuracy on 31,675 unseen flows

Random Forest classifies nine attack families

Gradient Boosting calibrates intrusion probability

Isolation Forest flags zero-day anomalies

SHAP shows the features behind each call

Retrieve

98.4%

Precision@1

Dense vectors in Qdrant fused with BM25 keyword match across 4,236 confirmed incidents.

Respond

AICredits → NVIDIA NIM → Ollama → offline playbooks

Automatic fallback, so an outage never blocks the analyst.

A risk score you can audit



Severity × confidence

How bad the family is, and how sure we are

Intrusion probability

Calibrated Gradient Boosting

Anomaly

Isolation Forest

Traffic signals

Rates, SYN, ports

Priority bands

P1
Critical
75+

P2
High
55+

P3
Medium
35+

P4
Low
below 35

Policy floors

Confident Heartbleed or Infiltration is always Critical. Botnet is never below High. Confirmed benign flows stay P4.

Built so analysts can trust it

Four guardrail layers

Schema bounds

IP, port and range checks on every flow

Injection filter

Screens text for adversarial prompts

Output structure

Enforces the four-section report format

Evidence check

Rejects incident IDs that were never retrieved

No containment without a human

Firewall blocks and host isolation are drafted, never executed. 0% unsafe actions measured.

Feedback loop

Analyst verdict → re-embed in Qdrant → retrain on drift

Every request is traced end to end with a trace ID.

Measured on CIC-IDS2017

99.0%

attack classifier accuracy

Macro-F1 0.952 across nine classes on unseen flows.

0.997

intrusion ROC-AUC

86.6% detection at 1% false positives

98.4%

retrieval Precision@1

Leave-one-out over 4,235 incidents

0%

hallucinations

Independent judge, 100% answer relevancy

Limits, and how to run it

Known limits

Rare classes

Heartbleed has 11 flows and Infiltration 36, so policy floors guarantee escalation.

Embedded Qdrant

Single process. Set QDRANT_URL for a distributed cluster.

Judge scores

LLM judges are signals, backed by deterministic guardrail tests.

Run it

```
setup.bat
start.bat
```

Run setup once to install everything, download CIC-IDS2017, train the models and build the index. Then start each day.

Open localhost:5173 and sign in as analyst, admin or viewer.